

DS-03 | KPI SPECIFICATION DOCUMENT

AI Insurance Claim Automation — UK NHS & Private Clinic Demo

Intern ID: DS-03	Week: W1 (5-9 May 2025)	Squad: Data Science	Status: Complete
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1. Objective

This document defines, justifies, and specifies all Key Performance Indicators (KPIs) to be displayed on the DS-03 KPI Dashboard throughout the AI Insurance Claim Automation project. These KPIs were identified in coordination with the Full Stack squad (FS) and UI/UX squad (UX) to ensure alignment between data availability, backend API design, and frontend display requirements.

The dashboard will serve as the primary reporting surface for NHS trust managers and clinic administrators to monitor claim processing performance in real time.

2. KPI Summary Table

#	KPI Name	Result	Status	Depends On
KPI 1	Approval Rate %	33.4%	Live	DS-03
KPI 2	Avg Days to Decision	Blocked	Blocked	FS-06
KPI 3	Fraud Flag Rate %	25.5%	Proxy	ML-03
KPI 4	Auto-Decision Rate %	11.6%	Proxy	ML-04
KPI 5	Total GBP Saved	96,437	Live	DS-03

3. KPI Definitions & Results

KPI 1 — Approval Rate %

Definition	Percentage of total claims with a final status of Paid
Formula	(Claim Status == 'Paid') / Total Claims x 100
Source Column	Claim Status
Dataset	claim_data.csv — 1,000 records
Result	33.4%
Status	Live — computed directly from dataset

KPI 2 — Average Days to Decision

Definition	Average number of calendar days between claim submission date and final decision date
Formula	Decision Date - Date of Service (average across all claims)
Source Column	Date of Service + Decision Date (missing from current schema)
Dataset	claim_data.csv — field not available
Result	BLOCKED
Status	Blocked — Decision Date field not present in current dataset. Escalated to FS-06 to add decision_date column to PostgreSQL schema. KPI will be computed once field is available in W3.

KPI 3 — Fraud Flag Rate %

Definition	Percentage of claims flagged as potentially fraudulent based on denial reason codes
Formula	$(\text{Reason Code IN ['Duplicate claim', 'Incorrect billing information']}) / \text{Total} \times 100$
Source Column	Reason Code
Dataset	claim_data.csv — 1,000 records
Result	25.5%
Status	Live (proxy) — Will be replaced with fraud_score field from ML-03 when available in W3

KPI 4 — Auto-Decision Rate %

Definition	Percentage of claims resolved automatically by the AI engine without requiring human review
Formula	$(\text{Outcome} == \text{'Paid'} \text{ AND Claim Status} == \text{'Paid'}) / \text{Total} \times 100$
Source Column	Outcome, Claim Status
Dataset	claim_data.csv — 1,000 records
Result	11.6%
Status	Live (proxy) — Will be replaced with auto_decision_flag from ML-04 rules engine when available

KPI 5 — Total GBP Saved

Definition	Total monetary difference between the amount billed and the amount actually paid across all claims — represents cost efficiency achieved through automated claim processing
Formula	$\text{SUM}(\text{Billed Amount} - \text{Paid Amount})$

Source Column	Billed Amount, Paid Amount
Dataset	claim_data.csv — 1,000 records
Result	Total: GBP 96,437 Avg per claim: GBP 96.44
Status	Live — computed directly from dataset

4. Data Gaps & Blockers

Gap	Impact	Blocked By	Action Taken
No decision_date field	KPI 2 cannot be computed	FS-06	Schema update requested — to be resolved in W3
No fraud_score field	KPI 3 uses proxy reason code logic	ML-03	Proxy in place — will update when /predict API is live in W3
No auto_decision_flag	KPI 4 uses proxy outcome logic	ML-04	Proxy in place — will update when rules engine ships in W3

5. Dataset Summary

File	claim_data.csv
Records	1,000
Date Range	May 2024 - September 2024
Key Columns	Claim Status, Reason Code, Billed Amount, Paid Amount, Outcome, Date of Service
Source	Synthetic dataset — generated by DS-01 for UK NHS demo purposes

6. Next Steps — W2

1. Build static Plotly/Dash dashboard displaying all 5 KPIs using synthetic data from claims_demo.csv
2. Display blocked KPIs (KPI 2) as placeholders with clear labels and blocker explanation
3. Coordinate with FS-06 on decision_date column addition to PostgreSQL schema
4. Coordinate with ML-03 on fraud_score field delivery timeline for W3 integration
5. Commit static dashboard prototype to GitHub before Friday 5 PM